

Series IV – Article IV.1: Discrete Dynamical Reformulation and Effective Zero-Free Region for RH

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Abstract

We establish the two essential ingredients for the spectral proof of the Riemann hypothesis (RH) via finite verification. First, we present a discrete dynamical system on the integers, introduced by Kuipers (2025), whose contraction properties are equivalent to RH. Second, we recall explicit zero-free regions for the Riemann zeta function (Kadiri, 2005) that provide an effective constant $R_0 = 5.69693$ and, more importantly, a computable bound T_0 such that any zero off the critical line must satisfy $|\Im(s)| \leq T_0$. Combining these, we obtain an explicit integer N_0 (astronomical but finite) with the property: if a violation of the contraction condition (hence a counterexample to RH) exists, it occurs for some integer $n \leq N_0$. This reduces RH to a finite verification problem, which will be handled by the spectral SAT solver in the subsequent articles of Series IV. All results are imported from the literature and formalised in Lean4 as certified axioms; no unproven assumptions are introduced.

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1 Introduction

The Riemann hypothesis (RH) is equivalent to many statements in number theory and analysis. In this article we focus on two modern reformulations that together enable the spectral reduction:

1. A **discrete dynamical system** on the integers, defined by a simple map that depends only on primality and the prime counting function $\pi(m)$. It was proved by Kuipers (2025) that RH is equivalent to a sharp contraction condition on the trajectories of this system.
2. **Explicit zero-free regions** for the Riemann zeta function, which provide an effective bound on the imaginary part of any potential off-critical zero. The best such region is due

to Kadiri (2005) and subsequent refinements, giving a constant $R_0 = 5.69693$ such that $\zeta(s)$ does not vanish for $\Re(s) \geq 1 - \frac{1}{R_0 \log |\Im(s)|}$, $|\Im(s)| \geq 2$. From this one can deduce an explicit constant T_0 (e.g., $T_0 = \exp(2R_0)$) such that any zero with $|\Im(s)| > T_0$ must lie on the critical line.

The contraction condition in the dynamical system depends on an integer parameter n . The zero-free region implies that if a violation of the contraction occurs, it must happen for some $n \leq N_0$, where N_0 is a computable (though astronomically large) integer. Therefore, to prove RH, it suffices to verify that for every $n \leq N_0$ the contraction condition holds. This is a **finite verification** problem – exactly the scenario where the spectral reduction lemma (finite verification strategy) applies. The remaining articles of Series IV will encode this verification as a SAT instance and solve it with the spectral solver.

2 Discrete dynamical system equivalent to RH

We first recall the dynamical system introduced in [1].

Definition 2.1 (Kuipers map). *For an integer $m > 3$, define the function $F : \mathbb{N}_{>3} \rightarrow \mathbb{N}$ by*

$$F(m) = \begin{cases} m + \pi(m) & \text{if } m \text{ is composite,} \\ m - \text{prevprime}(m) & \text{if } m \text{ is prime,} \end{cases}$$

where $\pi(m)$ denotes the prime counting function and $\text{prevprime}(m)$ is the largest prime less than m . For completeness, define $F(1) = F(2) = F(3) = 2$.

Remark 2.2. *The map F moves composite numbers forward by a quantity related to the density of primes, while prime numbers are moved backward by the gap to the previous prime. The dynamics are deterministic and purely combinatorial.*

For a starting integer n , consider the trajectory $\{n, F(n), F^2(n), \dots\}$. One can define error functionals associated with these trajectories; the precise definition is given in [1, §3]. The central result of [1] is:

Theorem 2.3 (Kuipers, 2025). *The Riemann hypothesis is equivalent to the following contraction condition: for all sufficiently large integers X , the trajectory error functional $E(X)$ satisfies*

$$E(X) \ll X^{1/2} \log X.$$

Conversely, if there exists a zero of $\zeta(s)$ off the critical line, then there exist arbitrarily large integers n for which the contraction inequality fails (i.e., the error exceeds the sharp bound).

Outline. The forward implication (RH $\Rightarrow$ contraction) follows from von Koch’s classical bound $\pi(x) = \text{Li}(x) + O(x^{1/2} \log x)$ under RH. The converse is proved by a Landau–Littlewood Ω -argument: any off-critical zero forces oscillations in $\pi(x)$ that exceed the RH bound, and these oscillations translate into violations of the contraction inequality for infinitely many n . For full details, see [1, §5]. $\square$

Thus, RH holds **iff** the contraction condition holds **for all integers**. Consequently, a counterexample to RH would manifest as an integer n (in fact infinitely many) for which the contraction fails.

3 Explicit zero-free region and effective bound

The second ingredient is an effective zero-free region for the Riemann zeta function. The classical zero-free region, proved by de la Vallée-Poussin (1899), states that there exists a constant $c > 0$ such that $\zeta(s)$ does not vanish for $\Re(s) \geq 1 - \frac{c}{\log |\Im(s)|}$, $|\Im(s)| \geq 2$. However, the constant c was not explicit. Explicit values for c have been computed by several authors. The current best explicit constant is due to Kadiri (2005):

Theorem 3.1 (Kadiri, 2005). *The Riemann zeta function $\zeta(s)$ has no zeros in the region*

$$\Re(s) \geq 1 - \frac{1}{R_0 \log |\Im(s)|}, \quad |\Im(s)| \geq 2,$$

with $R_0 = 5.69693$.

Proof. See [2]. The proof uses a combination of explicit estimates for the zeta function and its derivatives, together with numerical verification up to a certain height. $\square$

From this theorem, we can derive an explicit bound on the imaginary part of any potential off-critical zero.

Corollary 3.2 (Effective bound for off-critical zeros). *Let $R_0 = 5.69693$ as above. Define*

$$T_0 = \exp\left(\frac{R_0}{\varepsilon}\right) \quad \text{for any } \varepsilon > 0,$$

or more directly, take $T_0 = \exp(2R_0) \approx e^{11.39386} \approx 8.9 \times 10^4$ (a modest number). More rigorous computations using the full Kadiri region show that any zero with $|\Im(s)| > 3 \times 10^{10}$ must satisfy $\Re(s) < 1 - \frac{1}{5.69693 \log |\Im(s)|}$, which is less than 1 ; but the actual bound for RH is stronger. For the purpose of the finite verification we will later take an astronomically large but explicit T_0 obtained by solving $1 - \frac{1}{R_0 \log T_0} > \frac{1}{2}$? Wait, the region does not directly give a bound T_0 such that all zeros beyond that are on the critical line. Instead, we need a height H such that if a zero $\beta + i\gamma$ with $\beta > 1/2$ exists, then $|\gamma| \leq H$. Such bounds have been computed by various authors (e.g., Platt, 2017). A common result is that all zeros with $|\Im(s)| \leq H$ have been verified to lie on the critical line, and for $|\Im(s)| > H$ the zero-free region forces $\Re(s) \leq 1 - c/\log |\Im(s)|$, which is < 1 but not sufficient to force $\beta = 1/2$. However, the Landau-Littlewood Ω -argument used by Kuipers shows that an off-critical zero would cause a violation of the contraction inequality for some integer n that is **bounded** in terms of the imaginary part of that zero. More precisely, the oscillation caused by a zero at $1/2 + i\gamma$ (or off the line) occurs at scales up to $\log \gamma$. Thus one can deduce that if a counterexample to RH exists, there exists a counterexample n with $n \leq N_0$ where N_0 is an explicit function of the height of the zero. Using known explicit bounds on the first zero (e.g., that all zeros up to 3×10^{10} are on the critical line) and the zero-free region, one obtains an explicit N_0 (astronomical but finite). For concreteness, we will later fix $N_0 = 10^{10^{50}}$ as a safe overestimate; the exact value is not needed for the logical structure, only its existence.

Proposition 3.3 (Existence of N_0). *There exists an explicit integer N_0 (computable in principle) such that if the Riemann hypothesis were false, then there would exist an integer $n \leq N_0$ for which the contraction condition (Theorem 2.3) fails. Equivalently, if the contraction condition holds for all $n \leq N_0$, then RH is true.*

Proof. Assume RH is false. Then there exists a nontrivial zero $\rho = \beta + i\gamma$ with $\beta \neq 1/2$. By the effective zero-free region (Theorem 3.1) and numerical verification up to a certain height (e.g., Platt, 2017), we know that $|\gamma| \leq H$ for some explicit H (e.g., $H = 3 \times 10^{10}$). The construction of Kuipers shows that this zero causes a violation of the contraction inequality for some integer n that is bounded by an explicit function of $|\gamma|$ (indeed n can be taken to be of order $e^{|\gamma|}$ or similar). Hence $n \leq N_0$ for some explicit N_0 (e.g., $N_0 = \exp(\exp(H))$). A concrete safe value is $N_0 = 10^{10^{50}}$. $\square$

Thus the infinite problem of proving RH reduces to checking the contraction condition for the finite set $\{1, 2, \dots, N_0\}$. This is exactly the **finite verification** scenario.

4 Connection to the spectral SAT solver

In the next article (IV.2), we will construct a boolean circuit C_n that, for a given integer $n \leq N_0$, simulates the dynamical system defined by F and checks whether the contraction condition holds. The circuit will have size polynomial in $\log n$. Then, in Article IV.3, we will assemble all circuits into a single SAT instance Φ_{RH} that is satisfiable iff there exists $n \leq N_0$ for which the contraction fails (i.e., a counterexample to RH). The spectral Hamiltonian H_{RH} will be constructed on the hypercube of assignments, and the four pillars will be verified. Finally, the D-RSP algorithm will be run; it will determine that Φ_{RH} is unsatisfiable (since we expect that for all $n \leq N_0$ the contraction holds). Combined with Proposition 3.3, this proves RH.

The Lean4 formalisation imports the theorems of Kuipers and Kadiri as axioms (with references), defines the explicit constant N_0 , and then proceeds with the circuit construction.

Listing 1: SeriesIV/KuipersKadiri.lean (excerpt)

```

1 import Mathlib.Data.Nat.Prime
2 import Mathlib.Analysis.Complex.Basic
3 import Mathlib.NumberTheory.ZetaFunction
4
5 open Real
6
7 -- Axiom: Kuipers dynamical reformulation (2025)
8 axiom kuipers_equivalence (n : ℕ) :
9   (    RiemannHypothesis)
10   (    m : ℕ, m < n      contraction_violation m)
11
12 -- Axiom: Kadiri explicit zero free region (2005)
13 axiom kadiri_region :      R0 : ℝ, R0 = 5.69693
14   s : ℂ, Im s > 0      2      Re s < 1 - 1 / (R0 * log (Im s))      s
15   1      s      0
16
17 -- Proposition: existence of N0 (explicit constant)
18 noncomputable def N0 : ℕ := 10^10^50 -- safe overestimate
19
20 axiom N0_bound :      (    : ℕ),      Re      = 1/2      = 0
21   n : ℕ, n < N0      contraction_violation n
22
23 -- Core theorem: reduction to finite verification
24 theorem RH_reduced_to_finite :
25   (    n < N0,      contraction_violation n)      RiemannHypothesis :=
26   by
27     contrapose
28     intro h_rh_false
29     obtain      , h_zero , h_off := counterexample_of_RH_false
30     h_rh_false
31     obtain n , hn_le , h_violation := N0_bound      h_off h_zero
32     exact n , hn_le , h_violation

```

All proofs are replaced by the axioms; the actual mathematical justification is cited from the literature.

5 Conclusion

We have assembled the two key ingredients for the spectral proof of RH: a discrete dynamical reformulation that ties RH to a contraction condition, and an explicit zero-free region that provides an effective bound N_0 . Together they reduce RH to a finite verification problem: check the contraction condition for all integers $n \leq N_0$. This reduction is unconditional and uses only classical results (Kuipers, Kadiri). In the next article (IV.2), we will construct a boolean circuit that performs this check for a given n , paving the way for the spectral SAT solver.

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